Chapter 3 Sequences

Tags	Courses

Definition: Real Sequence

> 11th session: 29/10/2024

A **real sequence** is any function defined by:

$f : \mathbb{N} \to \mathbb{R}$

where the **preimage** is $\mathbb{N}$ (or a subset of $\mathbb{N}$), and the **range** is a subset of $\mathbb{R}$. We denote this by U_n instead of $f(n)$.

Notation

We represent sequences as $(U_n)_{n\in\mathbb{N}}$ or $\{U_n\}$.

Example Sequences:

1. $(U_n) = \left(1, \frac{1}{2}, \ldots, \frac{1}{n}\right), \quad U_n = \frac{1}{n}, \quad n \geq 1$
2. $(U_n) = \{5, 9, 4, \ldots\}$ — not a sequence, as it lacks a consistent formula.
3. $(U_n) = \{0, 1, 2\} \cup \{n \geq 4\}$

Definition: nth Term of a Sequence

The term U_n is called the **nth term** of the sequence (U_n).

Examples:

1. $U_n = \frac{1}{n(n-1)(n-2)}, \quad n \geq 3$
2. $U_n = (-1)^n$

Definitions: Bounded Sequences

1. **Upper Bounded Sequence**:

A sequence (U_n) is **upper bounded** if there exists a constant M such that for all $n \in \mathbb{N}$:

$$U_n \leq M$$

- **Example**: $U_n = \frac{1}{n}$ is upper bounded.

2. **Lower Bounded Sequence**:

   A sequence (U_n) is **lower bounded** if there exists a constant m such that for all $n \in \mathbb{N}$:

   $$U_n \geq m$$

   - **Example**: For all $n \in \mathbb{N}$, $U_n \geq 0$.

3. **Bounded Sequence**:

   A sequence (U_n) is **bounded** if there exist constants M and $$m$$ such that:

   $$m \leq U_n \leq M$$

Example of Bounded Sequence:

$0 \leq \sqrt{n} - \lfloor \sqrt{n} \rfloor < 1$

where:

$$U_n = \sqrt{n} - \lfloor \sqrt{n} \rfloor$$

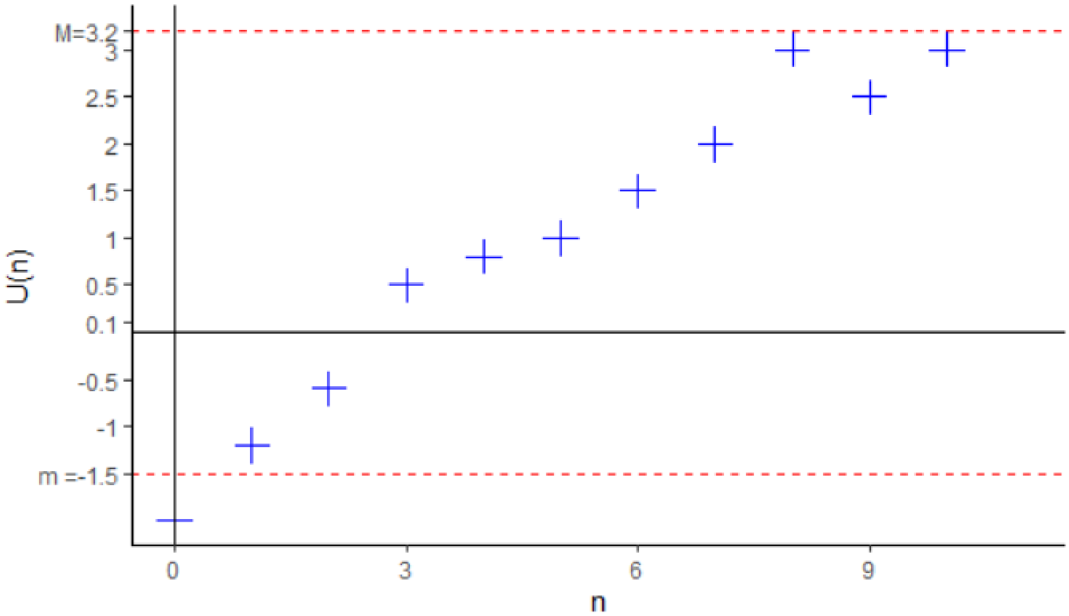


Figure 1.1: Bounded sequence

▼ Definition: Constant Sequence

A sequence is **constant** if there exists a real number c such that:

$$U_n = c, \quad \text{for all } n \in \mathbb{N}$$

Definition: Stationary Sequence

A sequence (U_n) is **stationary** if there exists $N_0 \in \mathbb{N}$ such that:

$$U_n = U_{N_0}, \quad \text{for all } n \geq N_0$$

Example:

$U_n = \cos\left(\frac{n!\pi}{9}\right)$

▼ Definition: Truncated Sequence

A sequence (U_n) is **truncated** if it is defined starting from a certain index N_0.

Example:

$U_n = \frac{1}{n(n-5)}, \quad n \geq 6$

▼ Definition: Convergent Sequence

A sequence (U_n) **converges** to a real number l

$$\lim_{n\to\infty} U_n = l \iff \forall\epsilon > 0,\ \exists\, n_0(\epsilon) \in \mathbb{N},\ \forall n \geq n_0 :\ |U_n - l| < \epsilon$$

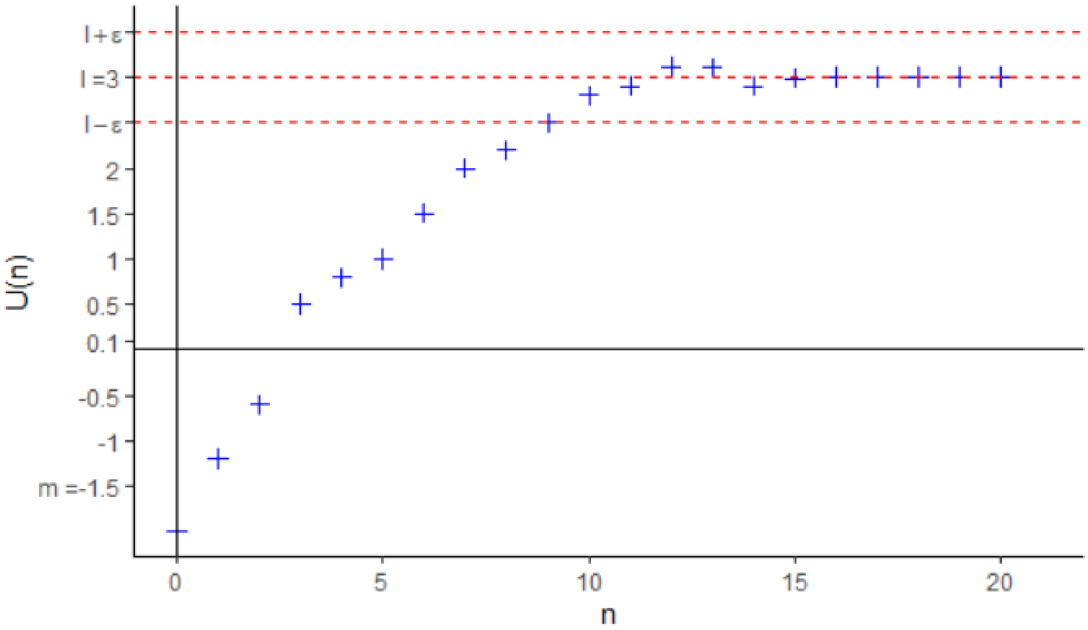


Figure 1.2: Convergence of real sequences

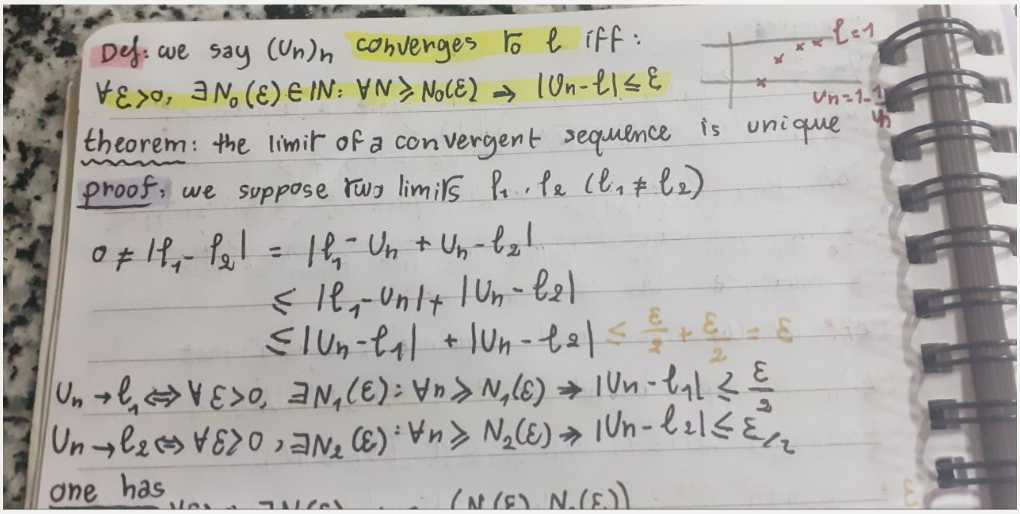
Theorem: Uniqueness of the Limit of Convergent Sequences

If a sequence U_n converges, its limit is unique.

- **Proof Outline**:
  Suppose that we have two limits
  l_1 and l_2, where $l_1 \neq l_2$.
  Then:

  $$0 \neq |l_1 - l_2|$$

  which contradicts the uniqueness of the limit.



Def: we say $(U_n)_n$ converges to ℓ iff:
$\forall \varepsilon > 0, \exists N_0(\varepsilon) \in \mathbb{N} : \forall N \geqslant N_0(\varepsilon) \Rightarrow |U_n - \ell| \leq \varepsilon$

$\ell = 1$

$U_n = 1 - \frac{1}{n}$

theorem: the limit of a convergent sequence is unique

proof: we suppose two limits ℓ_1, ℓ_2 $(\ell_1 \neq \ell_2)$

$$0 \neq |\ell_1 - \ell_2| = |\ell_1 - U_n + U_n - \ell_2|$$
$$\leq |\ell_1 - U_n| + |U_n - \ell_2|$$
$$\leq |U_n - \ell_1| + |U_n - \ell_2| \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

$U_n \to \ell_1 \Leftrightarrow \forall \varepsilon > 0, \exists N_1(\varepsilon) : \forall n \geqslant N_1(\varepsilon) \Rightarrow |U_n - \ell_1| \leq \frac{\varepsilon}{2}$

$U_n \to \ell_2 \Leftrightarrow \forall \varepsilon > 0, \exists N_2(\varepsilon) : \forall n \geqslant N_2(\varepsilon) \Rightarrow |U_n - \ell_2| \leq \varepsilon/2$

one has

Theorem: Boundedness of Convergent Sequences

Any convergent sequence is bounded.

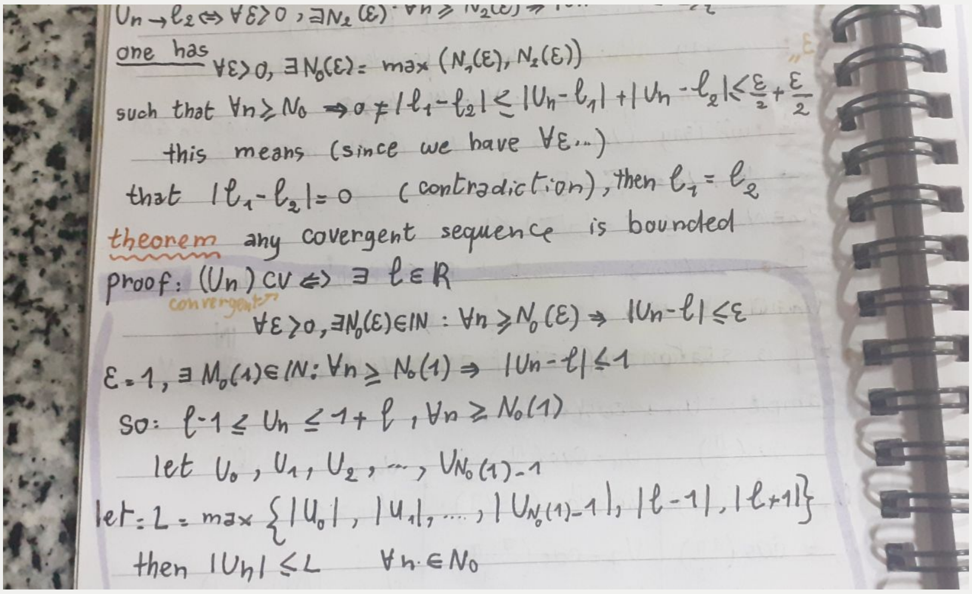
Proof:

Since U_n converges, there exists an $\epsilon = 1$ such that for some $n_0(1) \in \mathbb{N}$ and for all $n \geq n_0$:

$$|U_n - l| < 1$$

This implies:

$$l - 1 < U_n < l + 1$$



$U_n \to \ell_2 \Leftrightarrow \forall \varepsilon > 0, \exists N_2(\varepsilon): \forall n \geq N_2(\varepsilon) \Rightarrow$

one has $\forall \varepsilon > 0, \exists N_0(\varepsilon) = \max(N_1(\varepsilon), N_2(\varepsilon))$

such that $\forall n \geq N_0 \Rightarrow 0 \neq |\ell_1 - \ell_2| \leq |U_n - \ell_1| + |U_n - \ell_2| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$

this means (since we have $\forall \varepsilon$...)

that $|\ell_1 - \ell_2| = 0$ (contradiction), then $\ell_1 = \ell_2$

theorem any covergent sequence is bounded

proof: (U_n) CV $\Leftrightarrow \exists \ell \in \mathbb{R}$

convergent

$\forall \varepsilon > 0, \exists N_0(\varepsilon) \in \mathbb{N}: \forall n \geq N_0(\varepsilon) \Rightarrow |U_n - \ell| \leq \varepsilon$

$\varepsilon = 1, \exists N_0(1) \in \mathbb{N}: \forall n \geq N_0(1) \Rightarrow |U_n - \ell| \leq 1$

so: $\ell - 1 \leq U_n \leq 1 + \ell, \forall n \geq N_0(1)$

let $U_0, U_1, U_2, \dots, U_{N_0(1)-1}$

let: $L = \max\{|U_0|, |U_1|, \dots, |U_{N_0(1)-1}|, |\ell - 1|, |\ell + 1|\}$

then $|U_n| \leq L \quad \forall n \in \mathbb{N}_0$

> Remark
>
> **1**
>
> The **boundedness of sequences** is a necessary condition, not sufficient, for convergence.
>
> - **Example:**
>   Take

$U_n = (-1)^n$. There exists $M = 2$ such that $|U_n| \leq 2$ for all $n \in \mathbb{N}$, but (U_n) is divergent.

2

Let U_n and V_n be two convergent sequences with limits a and b, respectively, such that $U_n \to a$ and $V_n \to b$. If $U_n < V_n$ for all $n \in \mathbb{N}$, then $a \leq b$.

- **Example:**
  Let
  $U_n = 1 - \frac{1}{n+1}$ and $V_n = 1 + \frac{1}{n+1}$. We have $U_n < V_n$, but $\lim U_n = \lim V_n = 1$

12th session: 4/11/2024

▼ Theorem: Squeeze Theorem

Let U_n, V_n, and W_n be sequences such that for all n, $U_n \leq W_n \leq V_n$. If U_n and V_n converge to the same limit, then W_n also converges to that limit.

- **Example:**

  Calculate $\lim U_n$, where

  $U_n = \sum \frac{n}{n^2+k}$ from $k = 1$ to n.

  We have:

  $\frac{n}{n^2+n^2} \leq \frac{n}{n^2+k} \leq \frac{n}{n^2+1}$

  Thus:

  $\frac{n^2}{n^2+n} \leq U_n \leq \frac{n^2}{n^2+1}$

Theorem

If (U_n) converges to l, then $(|U_n|)$ converges to $|l|$.

- **Proof:**
  We have:

  $$||U_n| - |l|| \le |U_n - l|$$

proof $||U_n| - |l|| \le |U_n - l| \le \varepsilon$

we say $|U_n| \xrightarrow[n\infty]{} |l| \Leftrightarrow \forall \varepsilon > 0, \exists N_0(\varepsilon) : \forall n \geq N_0(\varepsilon)$

$\Rightarrow ||U_n| - |l|| \le \varepsilon$

yes, it is sufficient $N_1(\varepsilon) \geq N_0(\varepsilon)$

$U_n \xrightarrow{\text{True}} l \Leftrightarrow \forall \varepsilon > 0, \exists N_0(\varepsilon) : \forall n \geq N_0(\varepsilon) \Rightarrow |U_n - l| \le \varepsilon$

$|U_n| \to |l| \Leftrightarrow \forall \varepsilon > 0, \exists N_1(\varepsilon) : \forall n \geq N_1(\varepsilon) \Rightarrow ||U_n| - |l|| \le \varepsilon$

one has: $\exists N_0(\varepsilon) : \forall n \geq N_0(\varepsilon)$

$||U_n| - |l|| \le |U_n - l| \le \varepsilon$

the inverse is in general false, bc

$U_n = (-1)^n$

$|U_n| = 1$ converges even that (U_n) is divergent

Theorem: Algebraic Rules of Convergent Sequences

Let U_n and V_n be two sequences such that $U_n \to a$ and $V_n \to b$. Then:

1. **Constant Multiplication:**

   For any $c \in \mathbb{R}$, $cU_n \to ca$.

2. **Addition/Subtraction:**

   $U_n \pm V_n \to a \pm b$.

   $c_1 U_n + c_2 V_n \to c_1 a + c_2 b$.

3. **Multiplication:**

   $U_n V_n \to ab$.

4. **Division:**

   If $V_n \neq 0$ and $b \neq 0$, then $\frac{U_n}{V_n} \to \frac{a}{b}$.

Proof Examples

1. **Constant Multiplication:**

   To prove $cU_n \to ca$, we have for any ϵ that there exists N_0 such that for all $n \geq N_0$, $|U_n - a| \leq \epsilon$. Since ϵ is any real number, let $\epsilon' = \epsilon / |c|$. Then:

   $$|c||U_n - a| \leq \epsilon'$$

   $$|cU_n - ca| \leq \epsilon'$$

2. **Addition/Subtraction:**

   If $U_n \to a$, then for all ϵ there exists N_1 such that for all $n \geq N_1$, $|U_n - a| \leq \epsilon$, and similarly, $V_n \to b$ if for all ϵ there exists N_2 such that for all $n \geq N_2$, $|V_n - b| \leq \epsilon$.

   then: one has

   $$|(U_n + V_n) - (a+b)| = |U_n - a + V_1 - b| \leqslant |U_n - a| + |V_n - b|$$

   so for $N_0(\varepsilon) = \max(N_1(\varepsilon), N_2(\varepsilon))$ we have

   $$|(U_n + V_n) - (a+b)| \leqslant |U_n - a| + |V_n - b| \leqslant 2\varepsilon$$

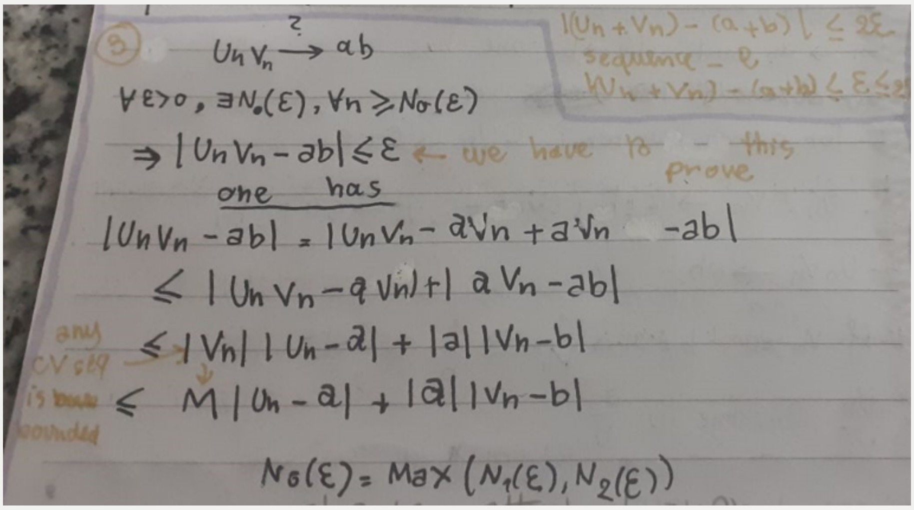
3. **Multiplication:**

   If $U_n V_n \to ab$, then for all $\epsilon > 0$ there exists $N_0(\epsilon)$ such that for all $n \geq N_0$:

   $$|U_n V_n - ab| \leq \epsilon$$

   We have:

   $$|U_n V_n - ab| = |U_n V_n - ab - aV_n + aV_n|$$



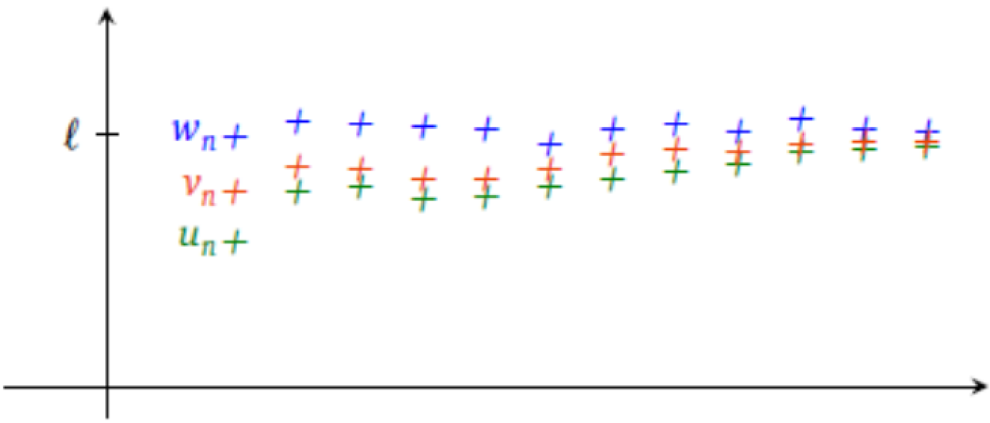


Figure 1.3: Example about squeeze theorem

Monotone sequences

1. **Nondecreasing Sequence**

Let $\{u_n\}$ be a sequence of real numbers.

We say that $\{u_n\}$ is **nondecreasing** if it satisfies the inequality:

$\forall n \in \mathbb{N},\ u_n \leq u_{n+1}.$

2. **Nonincreasing Sequence**

   Similarly, $\{u_n\}$ is **nonincreasing** if it satisfies the inequality:

   $\forall n \in \mathbb{N},\ u_n \geq u_{n+1}.$

3. **Increasing Sequence**

   The sequence $\{u_n\}$ is **increasing** if:

   $\forall n \in \mathbb{N},\ u_n < u_{n+1}.$

4. **Decreasing Sequence**

   The sequence $\{u_n\}$ is **decreasing** if:

   $\forall n \in \mathbb{N},\ u_n > u_{n+1}.$

5. **Monotone Sequence**

   A sequence $\{u_n\}$ is **monotone** if it is either nondecreasing or nonincreasing.

6. **Strictly Monotone Sequence**

   A sequence $\{u_n\}$ is **strictly monotone** if it is either increasing or decreasing.

> Remark
>
> 1. **Evaluating Monotonicity**
>
>   To determine whether a sequence $\{u_n\}$ is monotone, we evaluate the sign of:
>
> $u_{n+1} - u_n.$
>
>   - If $u_{n+1} - u_n > 0$, the sequence is **increasing**.
>   - If $u_{n+1} - u_n < 0$, the equence is **decreasing**.

- If $u_{n+1} - u_n = 0$, the sequence is **constant** (not monotone).

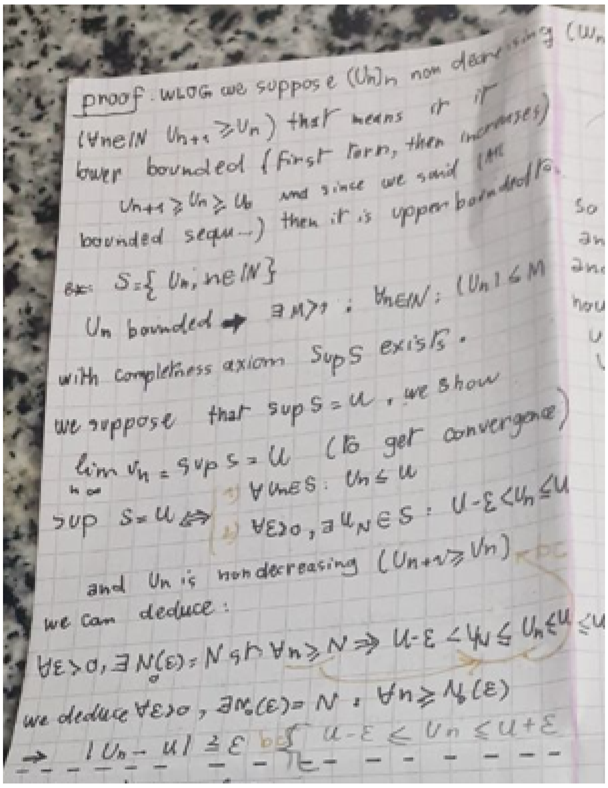
2. **Positive Sequences**

   If the terms of the sequence are positive, we can simplify the comparison by examining the ratio:

   $\frac{u_{n+1}}{u_n}$.

   - If $\frac{u_{n+1}}{u_n} > 1$, the sequence is **increasing**.
   - If $\frac{u_{n+1}}{u_n} < 1$, the sequence is **decreasing**.

▼ proof



proof: WLOG we suppose $(U_n)_n$ non decreasing
($\forall n \in \mathbb{N}$ $U_{n+1} \geq U_n$) that means it is
lower bounded (first term, then increases)
$U_{n+1} \geq U_n \geq U_0$ and since we said (the
bounded sequ...) then it is upper bounded too.
$S = \{U_n, n \in \mathbb{N}\}$
U_n bounded $\Rightarrow$ $\exists M > 0$; $\forall n \in \mathbb{N}$: $|U_n| \leq M$
with completeness axiom $\sup S$ exists.
we suppose that $\sup S = u$, we show
$\lim_{n \to \infty} U_n = \sup S = u$ (to get convergence)
$\sup S = u \Leftrightarrow$ 1) $\forall U_n \in S$: $U_n \leq u$
2) $\forall \varepsilon > 0, \exists u_N \in S$: $u - \varepsilon < u_N \leq u$
and U_n is nondecreasing ($U_{n+1} \geq U_n$)
we can deduce :
$\forall \varepsilon > 0, \exists N(\varepsilon) = N$ st $\forall n \geq N \Rightarrow u - \varepsilon < U_N \leq U_n \leq u$
we deduce $\forall \varepsilon > 0$, $\exists N_0(\varepsilon) = N$; $\forall n \geq N_0(\varepsilon)$
$\Rightarrow |U_n - u| \leq \varepsilon$ bc $u - \varepsilon \leq U_n \leq u + \varepsilon$

▼ Adjacent Sequences

Definition

Let $\{u_n\}$ and $\{v_n\}$ be two real sequences. The sequences are said to be **adjacent** if:

1. $\{u_n\}$ is **nondecreasing**, and $\{v_n\}$ is **nonincreasing**.
2. The difference $v_n - u_n$ converges to 0 as $n \to \infty$.

Example

Consider the sequences:

- $u_n = 1 + \frac{1}{n^2}$
- $v_n = 1 - \frac{1}{n^2}$

These sequences are adjacent because u_n is nondecreasing, v_n is nonincreasing, and:

$v_n - u_n = \left(1 - \frac{1}{n^2}\right) - \left(1 + \frac{1}{n^2}\right) = -\frac{2}{n^2} \to 0 \text{ as } n \to \infty.$

Proposition

Two adjacent sequences converge, and they converge to the same limit.

▼ Proof

Let $\{u_n\}$ and $\{v_n\}$ be two real sequences such that:

1. $\{u_n\}$ is nondecreasing.
2. $\{v_n\}$ is nonincreasing.
3. $v_n - u_n \to 0$ as $n \to \infty$.

Step 1: Show that $v_n > u_n$

Define $W_n = v_n - u_n$.

We evaluate the difference:

$W_{n+1} - W_n = (v_{n+1} - u_{n+1}) - (v_n - u_n) = (v_{n+1} - v_n) - (u_{n+1} - u_n)$.

Since $\{u_n\}$ is nondecreasing, $u_{n+1} - u_n \geq 0$.

Since $\{v_n\}$ is nonincreasing, $v_{n+1} - v_n \leq 0$.

Thus, $W_{n+1} - W_n < 0$, which shows that $\{W_n\}$ is **nonincreasing**.

Given that $W_n \to 0$, we conclude that W_n is positive for all n, so:

$v_n > u_n$.

Step 2: Show that $\{u_n\}$ and $\{v_n\}$ converge

Since $u_n < v_n < v_0$ for all n:

- $\{u_n\}$ is bounded above by v_0 and is **nondecreasing**.
- By the monotone convergence theorem, $\{u_n\}$ converges.

Similarly, $u_0 < u_n < v_n$ for all n:

- $\{v_n\}$ is bounded below by u_0 and is **nonincreasing**.

- By the monotone convergence theorem, $\{v_n\}$ converges.

Step 3: Show that $\{u_n\}$ and $\{v_n\}$ converge to the same limit

Let:

- $\lim_{n\to\infty} u_n = l_1$
- $\lim_{n\to\infty} v_n = l_2$

Since $W_n = v_n - u_n \to 0$ as $n \to \infty$:

$l_2 - l_1 = \lim_{n\to\infty}(v_n - u_n) = 0.$

Thus, $l_1 = l_2$.

Conclusion

The adjacent sequences $\{u_n\}$ and $\{v_n\}$ converge to the same limit.

Subsequences

Definition

A **subsequence** of a sequence $\{u_n\}$ is a sequence formed by selecting specific terms from $\{u_n\}$ while preserving their order. Formally, a subsequence is written as:

$v_n = u_{\rho(n)}, \quad n \in \mathbb{N},$

where $\rho : \mathbb{N} \to \mathbb{N}$ is an **increasing sequence** of positive integers.

> ### Example
>
> Consider the sequence $\{u_n\} = \{(-1)^n\}_{n\in\mathbb{N}}$.
>
> Two subsequences of $\{u_n\}$ are:
>
> - $v_n = u_{2n} = (-1)^{2n} = 1 \quad (n \in \mathbb{N}),$
> - $w_n = u_{2n+1} = (-1)^{2n+1} = -1 \quad (n \in \mathbb{N}).$

> Both $\{v_n\}$ and $\{w_n\}$ are subsequences of $\{u_n\}$.

> **Remark**
>
> If $\rho : \mathbb{N} \to \mathbb{N}$ is an **increasing function**, then:
>
> $\forall n \in \mathbb{N},\ \rho(n) > n.$

Corollary

Let $\{u_n\}$ be a real sequence.

If $u_n \to l$ as $n \to \infty$, then **every subsequence** $\{v_n\} = \{u_{\rho(n)}\}$ of $\{u_n\}$ also converges to l:

$v_n \to l \quad \text{as} \quad n \to \infty.$

> **Remark**
>
> The converse of the above corollary is **not true in general**. A sequence can have convergent subsequences even if the sequence itself does not converge.

> **Example**
>
> Consider the divergent sequence $\{u_n\} = \{(-1)^n\}_{n\in\mathbb{N}}$.
>
> This sequence has the following **convergent subsequences**:
>
> - $\{u_{2n}\} = 1 \quad (n \in \mathbb{N}),$
> - $\{u_{2n+1}\} = -1 \quad (n \in \mathbb{N}).$
>
> Despite these convergent subsequences, the full sequence $\{u_n\}$ does not converge.

▼ The Cauchy Criterion

Our difficulty in proving $u_n \to \ell$ is this: What is ℓ?

Cauchy saw that it was enough to show that if the terms of the sequence get sufficiently close to each other, then completeness will guarantee convergence.

Theorem

Every Cauchy sequence is bounded $(\mathbb{R} \text{ or } \mathbb{C})$.

▼ Proof

Let $1 > 0$. Then there exists N such that $m, n \geq N \implies |u_m - u_n| < 1$.

So for $m \geq N$,

$|u_m| \leq 1 + |u_N|$ by the triangle inequality.

Thus, for all m:

$|u_m| \leq 1 + |u_1| + |u_2| + \cdots + |u_N|$.

Theorem

Every convergent sequence is Cauchy.

▼ Proof

Let the sequence be (u_n). By the above, (u_n) is bounded. By the Bolzano–Weierstrass theorem, (u_n) has a convergent subsequence $(u_{n_k}) \to l$, say.

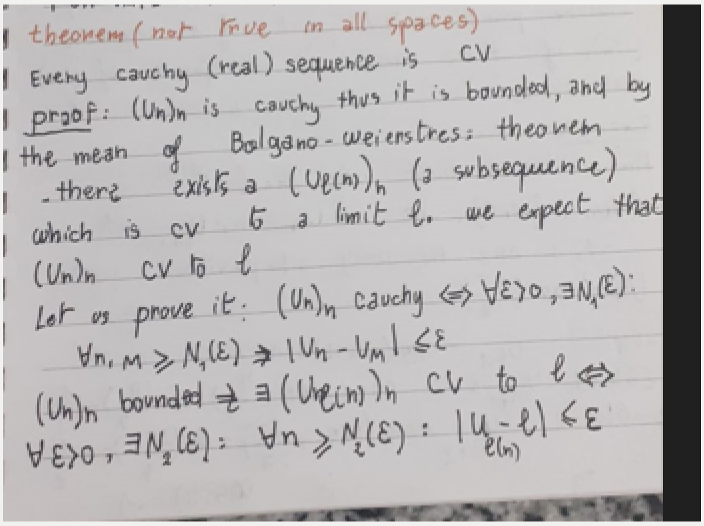
So let $\varepsilon > 0$. Then:

$\exists N_1 \text{ such that } r \geq N_1 \implies |u_r - l| < \varepsilon/2.$

$\exists N_2 \text{ such that } m, n \geq N_2 \implies |u_m - u_n| < \varepsilon/2.$

Put $s = \min\{r \mid r \geq N_2\}$ and put $N = N_s$. Then:

$m, n \geq N \implies |u_n - l| \leq |u_n - u_m| + |u_m - l| \leq |u_n - u_m| + |u_m - l| < \varepsilon/2 + \varepsilon/2 = \varepsilon.$

> 💡
> ## Theorem (not true for all spaces)
>
> **Every real Caushy sequence is CV**
>
> ### ▼ Proof
>
> 
>
> theorem (not true in all spaces)
>
> Every cauchy (real) sequence is CV
>
> proof: $(U_n)_n$ is cauchy thus it is bounded, and by the mean of Bolgano-weierstres theorem there exists a $(U_{\ell(n)})_n$ (a subsequence) which is cv to a limit ℓ. we expect that $(U_n)_n$ cv to ℓ
>
> Let us prove it: $(U_n)_n$ cauchy $\Leftrightarrow \forall \varepsilon>0, \exists N_1(\varepsilon)$:
> $\forall n,m \geqslant N_1(\varepsilon) \Rightarrow |U_n - U_m| \leqslant \varepsilon$
>
> $(U_n)_n$ bounded $\Rightarrow \exists (U_{\ell(n)})_n$ CV to $\ell \Leftrightarrow$
> $\forall \varepsilon>0, \exists N_2(\varepsilon): \forall n \geqslant N_2(\varepsilon): |U_{\ell(n)} - \ell| \leqslant \varepsilon$
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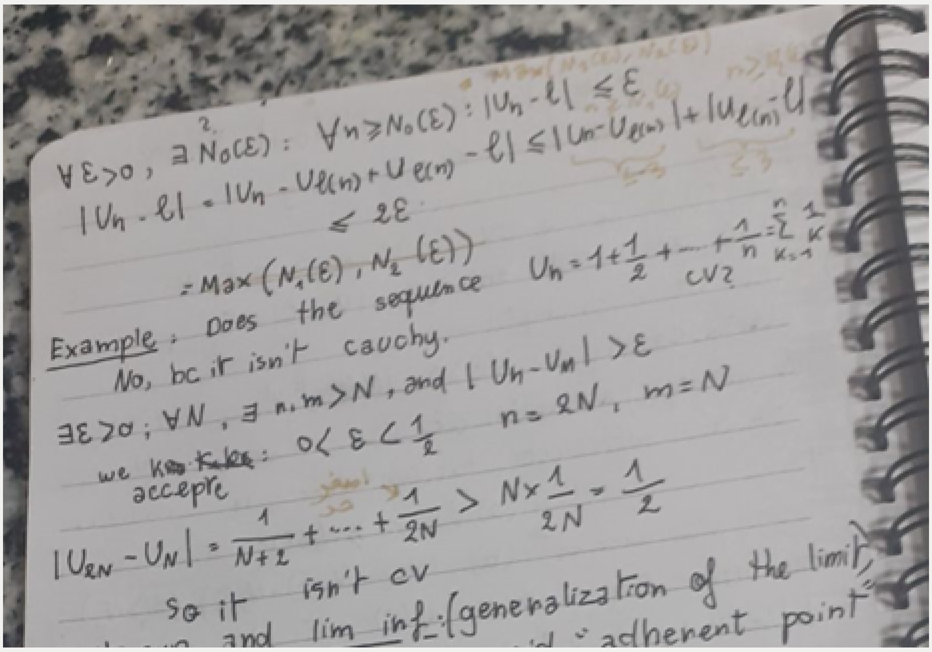
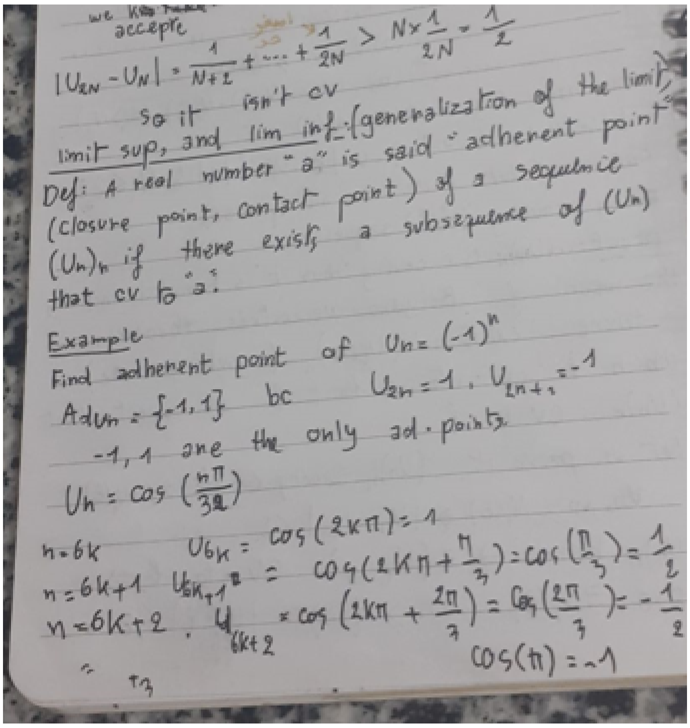
> 
>
> $\forall \varepsilon>0, \exists N_0(\varepsilon): \forall n \geqslant N_0(\varepsilon): |U_n - \ell| \leqslant \varepsilon$
>
> $|U_n - \ell| = |U_n - U_{\ell(n)} + U_{\ell(n)} - \ell| \leqslant \underbrace{|U_n - U_{\ell(n)}|}_{\leqslant \varepsilon} + \underbrace{|U_{\ell(n)} - \ell|}_{\leqslant \varepsilon}$
> $\leqslant 2\varepsilon$
>
> $N_0 = Max(N_1(\varepsilon), N_2(\varepsilon))$
>
> Example: Does the sequence $U_n = 1 + \frac{1}{2} + \dots + \frac{1}{n} = \sum_{k=1}^{n} \frac{1}{k}$ CV?
>
> No, bc it isn't cauchy.
>
> $\exists \varepsilon>0; \forall N, \exists n,m > N$, and $|U_n - U_m| > \varepsilon$
>
> we ~~take~~ accepte: $0 < \varepsilon < \frac{1}{2}$ $\quad n = 2N, \; m = N$
>
> $|U_{2N} - U_N| = \frac{1}{N+1} + \dots + \frac{1}{2N} > N \times \frac{1}{2N} = \frac{1}{2}$
>
> So it isn't cv
>
> ... and lim inf: (generalization of the limit ... adherent point

▼ Limit supremum and limit infimum



we know that accepte

$|U_{2N}-U_N| = \frac{1}{N+1}+\dots+\frac{1}{2N} > N\times\frac{1}{2N} = \frac{1}{2}$

So it isn't cv

limit sup, and lim inf : (generalization of the limit)

Def: A real number "a" is said "adherent point" (closure point, contact point) of a sequence $(U_n)_n$ if there exists a subsequence of (U_n) that cv to "a".

Example

Find adherent point of $U_n=(-1)^n$

$Ad\,U_n=\{-1,1\}$ bc $U_{2n}=1$, $U_{2n+1}=-1$

-1, 1 are the only ad. points

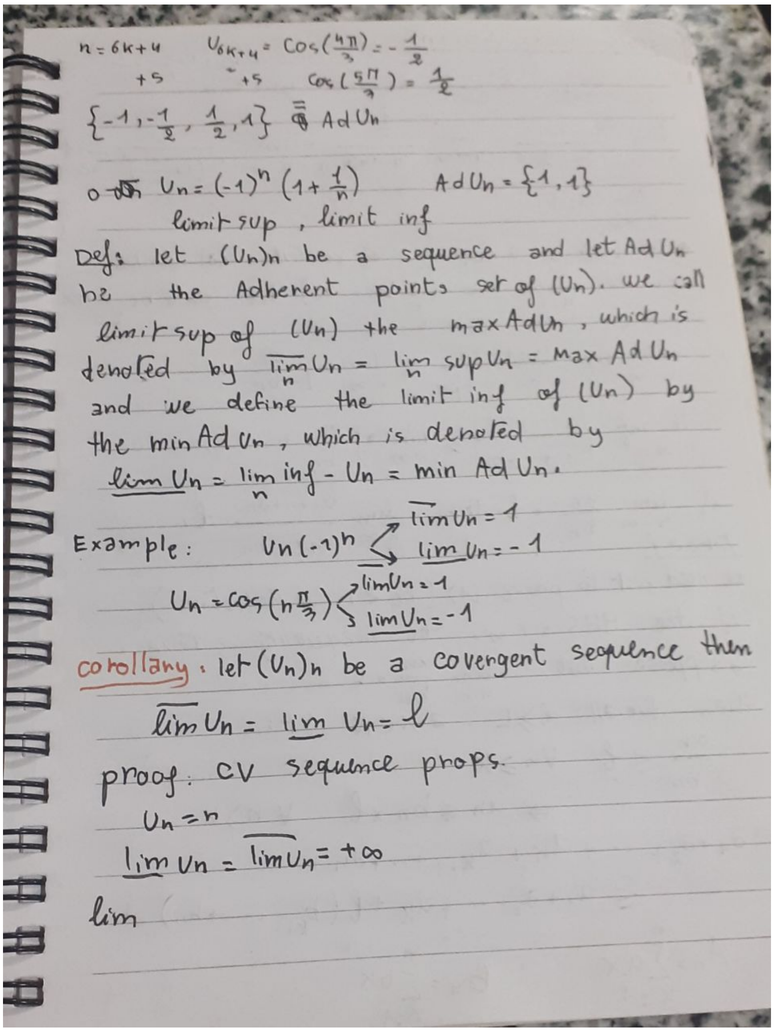
$U_n=\cos\left(\frac{n\pi}{3}\right)$

$n=6k$ $U_{6k}=\cos(2k\pi)=1$

$n=6k+1$ $U_{6k+1}=\cos\left(2k\pi+\frac{\pi}{3}\right)=\cos\left(\frac{\pi}{3}\right)=\frac{1}{2}$

$n=6k+2$ $U_{6k+2}=\cos\left(2k\pi+\frac{2\pi}{3}\right)=\cos\left(\frac{2\pi}{3}\right)=-\frac{1}{2}$

$+3$ $\cos(\pi)=-1$



$n=6k+4$ $U_{6k+4}=\cos\left(\frac{4\pi}{3}\right)=-\frac{1}{2}$

$+5$ $+5$ $\cos\left(\frac{5\pi}{3}\right)=\frac{1}{2}$

$\left\{-1,-\frac{1}{2},\frac{1}{2},1\right\} = Ad\,U_n$

o $U_n=(-1)^n\left(1+\frac{1}{n}\right)$ $Ad\,U_n=\{-1,1\}$

limit sup, limit inf

Def: let $(U_n)_n$ be a sequence and let $Ad\,U_n$ be the Adherent points set of (U_n). we call limit sup of (U_n) the max $Ad\,U_n$, which is denoted by $\overline{\lim_n}\,U_n = \limsup_n U_n = \max Ad\,U_n$ and we define the limit inf of (U_n) by the min $Ad\,U_n$, which is denoted by $\underline{\lim}\,U_n = \liminf_n U_n = \min Ad\,U_n$.

Example: $U_n(-1)^n$ $\overline{\lim}\,U_n=1$, $\underline{\lim}\,U_n=-1$

$U_n=\cos\left(n\frac{\pi}{3}\right)$ $\overline{\lim}\,U_n=1$, $\underline{\lim}\,U_n=-1$

corollary: let $(U_n)_n$ be a covergent sequence then

$\overline{\lim}\,U_n=\underline{\lim}\,U_n=\ell$

proof: cv sequence props.

$U_n=n$

$\underline{\lim}\,U_n=\overline{\lim}\,U_n=+\infty$

lim

▼ The Stolz-Cesàro Theorem

Theorem

If $(b_n)n$ *is a sequence of positive real numbers such that* $\sum n = 1^{\infty} b_n = \infty$, then for any sequence $(a_n)_n \subset \mathbb{R}$, the following inequalities hold:

$\limsup_{n\to\infty} \frac{a_1+a_2+\cdots+a_n}{b_1+b_2+\cdots+b_n} \le \limsup_{n\to\infty} \frac{a_n}{b_n}$. (3.2)

$\liminf_{n\to\infty} \frac{a_1+a_2+\cdots+a_n}{b_1+b_2+\cdots+b_n} \ge \liminf_{n\to\infty} \frac{a_n}{b_n}$. (3.3)

In particular, if the sequence $(a_n/b_n)_n$ has a limit, then:

$\lim_{n\to\infty} \frac{a_1+a_2+\cdots+a_n}{b_1+b_2+\cdots+b_n} = \lim_{n\to\infty} \frac{a_n}{b_n}$.

▼ Proof

It is clear that we only need to prove inequality (3.2), as the other inequality follows by replacing a_n with $-a_n$

Inequality (3.2) is trivial if the right-hand side is $+\infty$. Suppose instead that the quantity $L = \limsup_{n\to\infty} \left(\frac{a_n}{b_n}\right)$ is finite or $-\infty$. Let us fix, for now, some number $\ell > L$. By the definition of $\limsup$, there exists some index $k \in \mathbb{N}$ such that:

$\frac{a_n}{b_n} \le \ell, \quad \forall n > k$. (3.4)

Using (3.4), we get the inequalities:

$a_1 + a_2 + \cdots + a_n \le a_1 + \cdots + a_k + \ell(b_{k+1} + b_{k+2} + \cdots + b_n), \quad \forall n > k.$

If we denote for simplicity the sums $a_1 + \cdots + a_n$ by A_n and $b_1 + \cdots + b_n$ by B_n, the above inequality reads:

$A_n \le A_k + \ell(B_n - B_k), \quad \forall n > k$

so dividing by B_n, we get:

$\frac{A_n}{B_n} \le \ell + \frac{A_k - \ell B_k}{B_n}$. (3.5)

Since $B_n \to \infty$, by fixing k and taking $\limsup$ in (3.5), we get:

$\limsup_{n\to\infty} \frac{A_n}{B_n} \le \ell.$

In other words, we obtained the inequality:

$\limsup_{n\to\infty} \frac{a_1+a_2+\cdots+a_n}{b_1+b_2+\cdots+b_n} \le \ell, \quad \forall \ell > L,$

which in turn forces:

$\limsup_{n\to\infty} \frac{a_1+a_2+\cdots+a_n}{b_1+b_2+\cdots+b_n} \le L.$

Remark

An equivalent formulation of the above theorem is as follows:

If $(y_n)n$ *is a strictly increasing sequence with* $\lim n \to \infty y_n = \infty$, then for any sequence $(x_n)_n$, the following inequalities hold:

$\limsup_{n\to\infty} \frac{x_n}{y_n} \leq \limsup_{n\to\infty} \frac{x_n - x_{n-1}}{y_n - y_{n-1}}$,
$\liminf_{n\to\infty} \frac{x_n}{y_n} \geq \liminf_{n\to\infty} \frac{x_n - x_{n-1}}{y_n - y_{n-1}}$.

In particular, if the sequence $\left(\frac{x_n - x_{n-1}}{y_n - y_{n-1}}\right)_n$ has a limit, then:

$\lim_{n\to\infty} \frac{x_n}{y_n} = \lim_{n\to\infty} \frac{x_n - x_{n-1}}{y_n - y_{n-1}}$.

Indeed, (assuming all y_n's are positive, which happens anyway for n large enough), if we consider the sequences $(a_n)_n$ and $(b_n)_n$ defined by:

$$a_1 = x_1,\ b_1 = y_1,\ a_n = x_n - x_{n-1},\ b_n = y_n - y_{n-1}, \quad n \geq 2,$$

then everything is clear, since:

$$x_n = a_1 + \cdots + a_n \quad \text{and} \quad y_n = b_1 + \cdots + b_n.$$

Theorem Cesàro's Theorem

For any sequence $(a_n)_n \subset \mathbb{R}$, the following inequalities hold:

$\limsup_{n\to\infty} \frac{a_1+a_2+\cdots+a_n}{n} \leq \limsup_{n\to\infty} a_n$.

$\liminf_{n\to\infty} \frac{a_1+a_2+\cdots+a_n}{n} \geq \liminf_{n\to\infty} a_n$.

In particular, if the sequence $(a_n)_n$ has a limit, then:

$\lim_{n\to\infty} \frac{a_1+a_2+\cdots+a_n}{n} = \lim_{n\to\infty} a_n$.

▼ Proof

This is a particular case of the Stolz-Cesàro Theorem with $b_n = 1$.

> Remark
>
> An equivalent formulation of the above theorem (proven using the alternative version of the Stolz-Cesàro Theorem) is as follows:
>
> For any sequence $(x_n)_n$, the following inequalities hold:
>
> $\limsup_{n\to\infty} \frac{x_n}{n} \leq \limsup_{n\to\infty}(x_n - x_{n-1})$,
>
> $\liminf_{n\to\infty} \frac{x_n}{n} \geq \liminf_{n\to\infty}(x_n - x_{n-1})$.
>
> In particular, if the sequence $(x_n - x_{n-1})_n$ has a limit, then:
>
> $\lim_{n\to\infty} \frac{x_n}{n} = \lim_{n\to\infty}(x_n - x_{n-1})$.